Fiscal Year:	FY 2011	**Task Last Updated:**	FY 12/09/2011
PI Name:	Cartreine, James A. Ph.D.		
Project Title:	Countermeasure for Managing Interpersonal Conflicts in Space: A Continuation Study		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jcartreine@partners.org	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-851-8913
Organization Name:	Brigham and Women's Hospital/Harvard Medical Center		
PI Address 1:	Program in Behavioral Informatics and eHealth		
PI Address 2:	Department of Psychiatry		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:	NOTE: PI moved to Brigham and Women's Hospital; formerly at Beth Israel Deaconess Medical Center, Boston, MA, per info received December 2011 (Ed.) Name change to Cartreine in summer 2008 (from Carter), per NSBRI information (11/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	09/01/2009	**End Date:**	10/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Greenhalgh, Leonard (Dartmouth College) Beven, Gary (NASA Johnson Space Center)		
Grant/Contract No.:	NCC 9-58-NBPF01603		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Some amount of interpersonal conflict is expected on long-duration (LD) missions, whether between crewmembers or between the crew and the ground; however, severe conflicts can interfere with mission success and even safety. Severe, ongoing conflicts have been reported on LD space missions, Antarctic expeditions, and a recent LD space mission simulation. Nonetheless, empirically-supported tools to help crews prevent, assess, and manage interpersonal conflicts have not been developed. This study addresses Bioastronautics Roadmap Risk #24: Human Performance Failure Due to Poor Psychosocial Adaptation. The investigators have begun development of an interactive multimedia portal to help LD flyers prevent, detect, assess, and manage their own psychosocial problems. This work has been supported by two NSBRI grants. In the first grant, program architecture was developed to organize the types of content and experiences users can find in the portal. Sample content about depression and interpersonal conflict was also developed as a prototype to demonstrate how training could be delivered through the system. The second NSBRI grant was to complete and evaluate the depression content, including self-treatment of depression. The present project will design and produce an interactive media intervention program to assist persons manage real, ongoing conflicts on long-duration missions. A preliminary evaluation of the conflict intervention will be conducted in a variety of settings where individuals live, train, and work together: fire departments. The goal of this preliminary evaluation is to gather data on response rate, sample characteristics, and to estimate effect size, in preparation for a randomized controlled trial. ORIGINAL AIMS: 1. The primary deliverable will be a countermeasure to help crews manage interpersonal conflict. We will produce an interactive media intervention program to assist persons manage real, ongoing conflicts on LD missions. This intervention will be based on cognitive-behavioral therapy and will help individuals to work out strategies with the assistance of a coach, on computer. 2. A second deliverable will be preliminary data about the countermeasure. We will conduct an open trial to obtain data about the intervention's potential effectiveness in settings that share some characteristics of the space setting (group living, small work groups). The open trial will also enable us to develop and assess our procedures and instruments for data collection, and estimate effect size, in preparation for a future randomized controlled trial. KEY FINDINGS: Year 2 has involved 5 major activities: 1. Writing the Design Document for the intervention program, 2. Engineering the process of data collection and transfer for the program trial, 3. Securing IRB approval, 4. Writing the storyboard and shooting script for the first 2 (of 4) sessions, 5. Programming the first session. Additionally, we have continued to build trust and rapport, and do evaluation planning with the Houston Fire Department, the Phoenix Fire Department, and the San Diego Fire Department. The program will be evaluated at these sites. The year has been tremendously busy and productive on all of the above fronts; each of which have posed their unique academic, intellectual, organizational and administrative challenges. PROPOSED RESEARCH PLAN: In Year 3 we will complete the production of the conflict intervention program, conduct alpha testing (debugging and in-house usability testing) and begin the evaluation of the program in the analog settings of real fire departments with fire fighters who are involved in real, ongoing interpersonal conflicts. Test sites will be the Houston Fire Department, the Phoenix Fire Department and the San Diego Fire Department. In Year 4, we will complete the evaluation study in those departments, making any necessary revisions to the software as we go. Also in Year 4 we will prepare a paper for publication about the program and its use in fire departments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Clearly, if effective, this countermeasure could help firefighters and other first responders. It could also be adapted for use in isolated operational environments, such as polar research stations, submarines, commercial ships, oilrigs, and underwater research stations. Furthermore, even greater value could be derived by making similar programs available to the public in other settings such as police departments, secondary schools, social services offices, places of worship, military bases, prisons, public health and mental health centers, and eventually at home or in any location, through broadband Internet.
Task Progress:	As noted in Main Findings, above, Year 2 has involved 5 major activities: 1. Writing the Design Document for the intervention program, 2. Engineering the process of data collection and transfer for the program trial, 3. Securing IRB approval, 4. Writing the storyboard and shooting script for the first 2 (of 4) sessions, 5. Programming the first session. Additionally, we have continued to build trust and rapport, and do evaluation planning with the Houston Fire Department, the Phoenix Fire Department, and the San Diego Fire Department. The program will be evaluated at these sites. Through this year we have learned that: 1. It is possible to implement the conceived research plan, from a technical perspective: the intervention is delivered via USB flashdrive, data is uploaded at the end of each session to a cloud computer that is controlled by Partners Healthcare (which owns Brigham and Women's Hospital). All data will be protected to HIPAA-level standards, although no protected health information (PHI) will actually be transmitted. 2. It is possible to design the program in such a way that it tailors the intervention to the user's unique situation; providing guidance and training they need based on an extensive algorithm. Users will answer questions derived from standardized measures and this data will be fed to the algorithm to determine the user's intervention plan. It should be noted that no previous intervention for workplace conflict has ever been empirically evaluated in any workplace. Therefore, no evidence-based intervention was available for "off the shelf" adaptation. The investigators created an intervention plan that is based on the literature from couples therapy, mediation, workplace conflict epidemiology (i.e., studies of what typically happens in workplace conflict, versus interventions to manage it), and school-based conflict interventions. Our program appears to be the first intervention for workplace conflict that will be evaluated in a scientific manner.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Articles in Peer-reviewed Journals	Cartreine JA, Locke SE, Buckey JC, Sandoval L, Hegel M. "Interactive Media Treatment for Depression: Description and Feasibility Evaluation." Journal of Medical Internet Research. In press, August 2011. , Aug-2011